Apply filters to SQL queries

Project description

This report was written to demonstrate the knowledge acquired about SQL throughout the fourth Google cybersecurity course. This demonstration will be done by simulating an investigation into an organization's security issues in order to help keep its systems safer.

The task is to examine the organization's data in the `employees` and `log_in_attempts` tables. To do so, SQL filters will be used to retrieve records from different datasets and investigate potential security issues.

Retrieve after hours failed login attempts

The first vulnerability that could lead to a security incident was identified after business hours.

To investigate this, a query was used in the `log_in_attempts` table to review login activity that occurred after 18:00 and included failed attempts. This was performed using the following query.

```
MariaDB [organization]> SELECT *
    -> FROM log_in_attempts
    -> WHERE login_time > '18:00:00' AND success = 0;
```

Figure 01: Reviewing login activity querry

This query returns all columns from the `log_in_attempts` table where `login_time`, the column that stores the system login time, is greater than `18:00:00`, and where `success` indicates failed login attempts. Failed attempts are represented by `0` in binary.

Retrieve login attempts on specific dates

During the investigation, it was discovered that a suspicious event occurred on `2022-05-09`. Therefore, it is necessary to review all login attempts that occurred on that day and the day before. This task can be seen in the following query.

```
MariaDB [organization]> SELECT *
    -> FROM log_in_attempts
    -> WHERE login_date = '2022-05-09' OR login_date = '2022-05-08';
```

Figure 02: Reviewing all suspicious events on the specific day

The login date is stored in the `login_date` column. This query returns all columns from the `log_in_attempts` table where `login_date` is equal to either the suspicious event date, `2022-05-09`, or the previous day, `2022-05-08`.

Retrieve login attempts outside of Mexico

There has been suspicious activity involving login attempts, but it was discovered that this activity did not originate from Mexico. Therefore, it is necessary to retrieve the login attempts that occurred outside of Mexico. This task is performed using the query below.

```
MariaDB [organization]> SELECT *
    -> FROM log_in_attempts
    -> WHERE NOT country LIKE 'MEX%';
```

Figure 03: Retrieving login attempts that occurred outside of Mexico

The `NOT` operator, used together with `WHERE`, tells SQL to return the countries stored in the `country` column that do not start with `MEX`. In the table, Mexico may be stored as `MEX` or `MEXICO`. The `LIKE` keyword together with `%` ensures that the query identifies any country value that starts with `MEX`.

Retrieve employees in Marketing

The team decided to perform security updates on specific employee machines. Their targets are employees who belong to the Marketing department and are located in the East offices.

The employee department is stored in the `department` column, and the office location is stored in the `office` column. Keep in mind that there are different offices located on the East side.

```
MariaDB [organization]> SELECT *
    -> FROM employees
    -> WHERE department = 'Marketing' AND office LIKE 'East%';
```

Figure 04: Reviewing all employees based on department and office

This query returns all columns from the `employees` table, but only for employees whose `department` column is equal to `Marketing` and whose `office` starts with `East`.

Retrieve employees in Finance or Sales

Now, it is necessary to perform more security updates, but this time on machines belonging to employees in the Sales and Finance departments. The query remains almost the same as the previous one, with only a small change to instruct SQL to return all employees who belong to either the Sales or Finance departments.

```
MariaDB [organization]> SELECT *
    -> FROM employees
    -> WHERE department = 'Sales' OR department = 'Finance';
```

Figure 05: Retrieving employees in Finance or Sales

Retrieve all employees not in IT

For the last security update, it is necessary to update the employees' machines. It is known that only the employees in the Information Technology department have already received this update, meaning that employees from all other departments still need it.

To solve this, a query was written to identify all employees who are not in the IT department. In other words, it filters employees whose `department` column does not contain `Information Technology`.

```
MariaDB [organization]> SELECT *
    -> FROM employees
    -> WHERE NOT department = 'Information Technology';
```

Figure 06: Retrieving all employees not in IT

Summary

The purpose of this report was to develop and improve knowledge of SQL filters and operators, such as equals (`=`), greater than (`>`), and less than (`<`), as well as logical operators like `AND`, `OR`, and `NOT`.

This knowledge is extremely important in the security analyst field, since security professionals may encounter tasks involving security issues or incidents that require the use of relational databases.